

A 1.30^d Lower Bound for the Chromatic Number of Euclidean Space

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THE RESULT

ORDINARY UNIT-DISTANCE COLOURING

For an absolute constant $c > 0$ and all sufficiently large d ,

$$\chi(\mathbb{R}^d) \geq c C_*^d, \quad 1.309251 < C_* < 1.309252.$$

In particular, $\chi(\mathbb{R}^d) \geq 1.30^d$ eventually. For infinitely many dimensions, the same construction gives $\chi(\mathbb{R}^d) \geq 1.316^d$.

01 The geometric idea

For $Q = 6$ or 7 , let $\mathcal{A}_Q$ consist of the first Q entries of $(0, 1, -1, 2, -2, 3, -3)$. For a power of two $P \geq 2$, take the entire layer

$$V = \left\{ x \in \mathcal{A}_Q^n : \sum_{i=1}^n (2x_i^2 - x_i) = 2P - 1 \right\}.$$

Its points lie on a sphere about $o = (1/4, \dots, 1/4)$. Their midpoints lie in $(\frac{1}{2}\mathbb{Z})^n$, at squared distance at least $n/16$ from o . The parallelogram identity and parity give

$$\rho(x, y) := \frac{1}{2}\|x - y\|^2 \in \{0, \dots, 2P - 1\}.$$

Joining pairs with $\rho = P$ gives a unit-distance graph after scaling.

02 From geometry to a rank bound

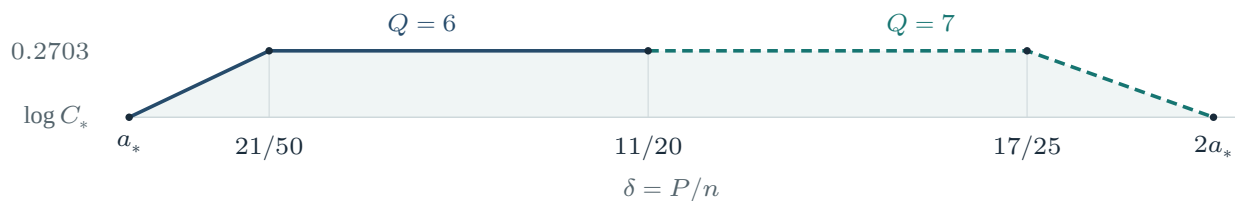
Over $\mathbb{F}_2$, the matrix $\mathcal{M}_{xy} = \binom{\rho(x,y)-1}{P-1}$ is one exactly on the diagonal and the edges, with $\binom{-1}{P-1} = (-1)^{P-1}$. Every independent set therefore indexes an identity matrix. Newton interpolation exposes powers of a formal variable; coefficient extraction then reduces the rank estimate to a sum of outer products. For $V \neq \emptyset$, this yields

$$\chi(\mathbb{R}^n) \geq \frac{|V|}{\text{rank}_{\mathbb{F}_2} \mathcal{M}} \geq \frac{[t^{2P-1}]A_Q(t)^n}{[s^{P-1}](E_Q(s)^n/(1-s)^2)}.$$

Here $A_Q(t) = \sum_{0 \leq j < Q} t^{j(j+1)/2}$, $E_Q(s) = \sum_{0 \leq j < Q} s^{e_j}$, and $(e_0, \dots, e_6) = (0, 1, 3, 4, 8, 9, 11)$; brackets denote coefficient extraction.

03 A constant prefactor, uniformly in the dimension

An elementary binomial estimate gives matching $n^{-1/2}$ factors in the vertex count and rank bound; these cancel. The least power of two $P \geq a_* n$, where $a_* \approx 0.37198$, gives $P/n \in [a_*, 2a_*)$. Four concave lower bounds for the logarithmic rate cover this interval using six or seven coordinate values.



The concave lower bounds lie above the four indicated chords.

Four exact integer witnesses with common frequency denominator $M = 200$ give a short route to 1.30^d . The full paper includes all algebraic proofs and finite arithmetic data.